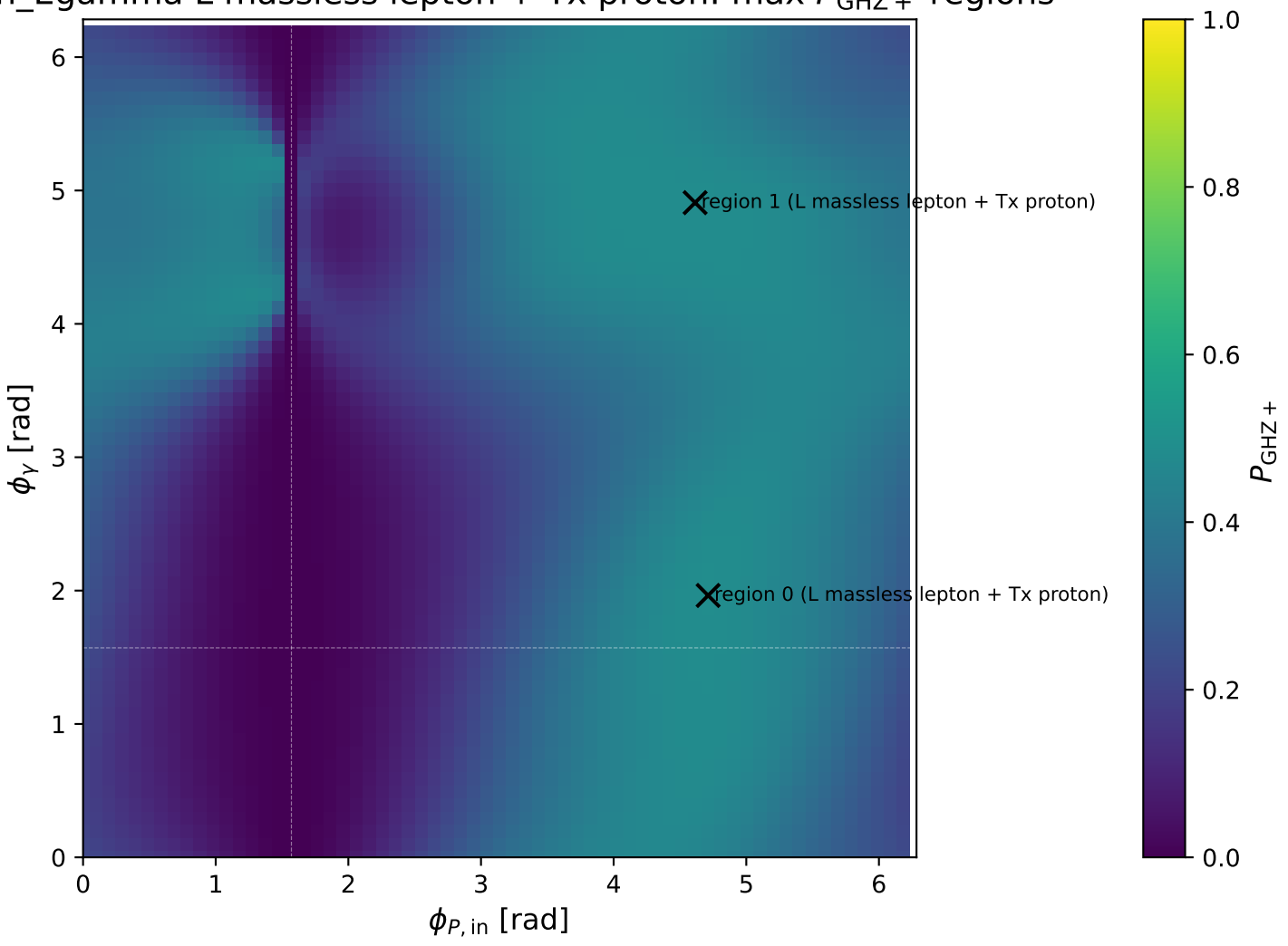
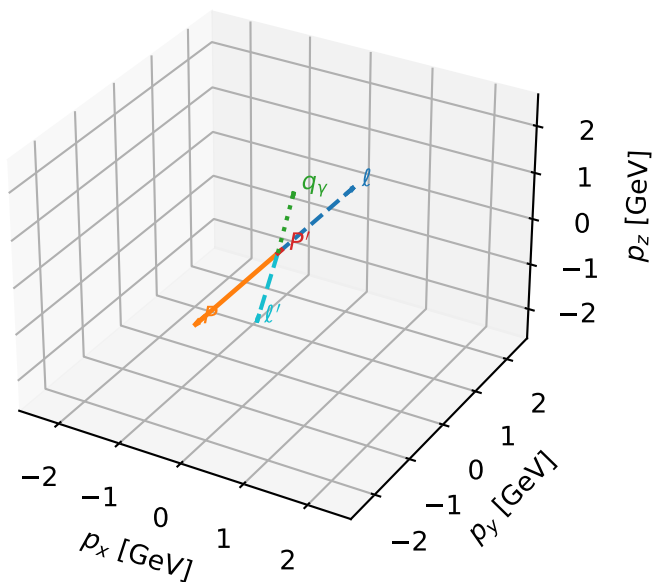


# $\gamma$ L massless lepton + Tx proton: max $P_{\text{GHZ}}$ + regions



# L massless lepton + Tx proton characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

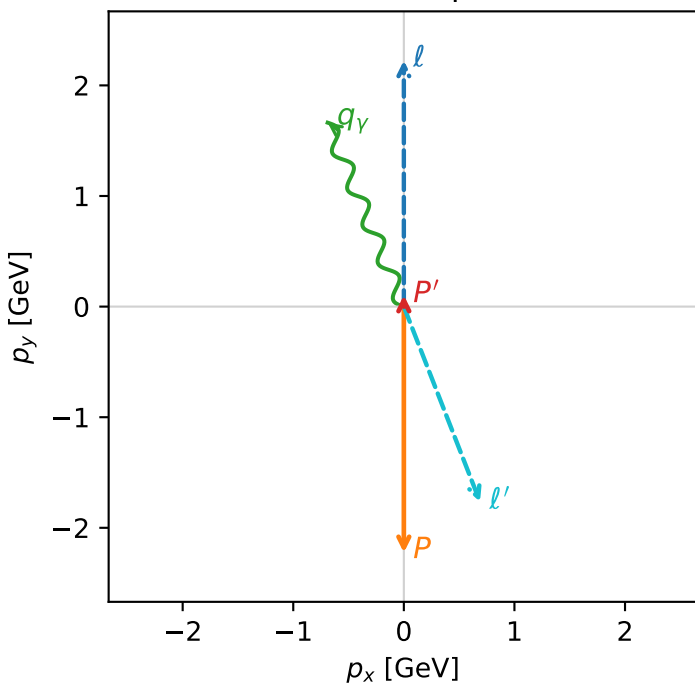


ghz\_purity\_high\_egamma\_ltx\_region\_0  $P_{\text{GHZ}}=0.485135$   
 region: high\_Egamma\_LTx\_region\_0  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|L_{l_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=0.102337$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_l=1.5708$ ,  $\phi_p=4.71239$   
 $E_\gamma=1.8$ ,  $\phi_\gamma=1.9635$   
 $Q^2=16.2839$ ,  $x_B=0.841965$ ,  $t=-3.25132$ ,  $W^2=3.93631$ ,  $y=0.937217$   
 $\theta(l', \gamma)=178.816$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 0.0000$   $p_y= 2.2244$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.0000$   $p_y= -2.2244$   $p_z= 0.0000$   
 $\ell'$   $E= 1.8950$   $p_x= 0.6888$   $p_y= -1.7653$   $p_z= 0.0000$   
 $P'$   $E= 0.9436$   $p_x= 0.0000$   $p_y= 0.1023$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= -0.6888$   $p_y= 1.6630$   $p_z= 0.0000$

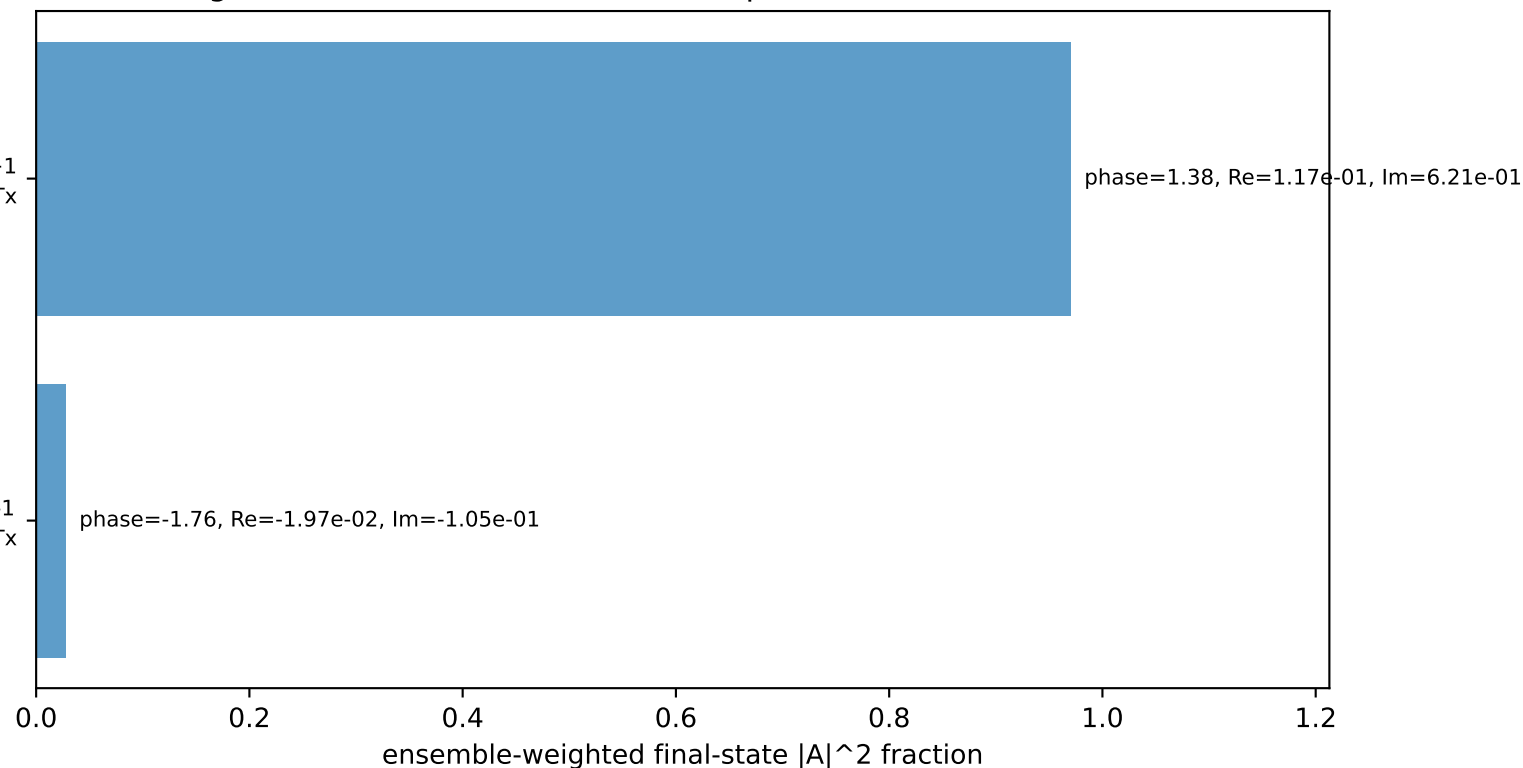
Transverse plane



$h_e=+1$ ,  $h_p=+1$ ,  $h_\gamma=+1$   
 massless lepton L, proton Tx

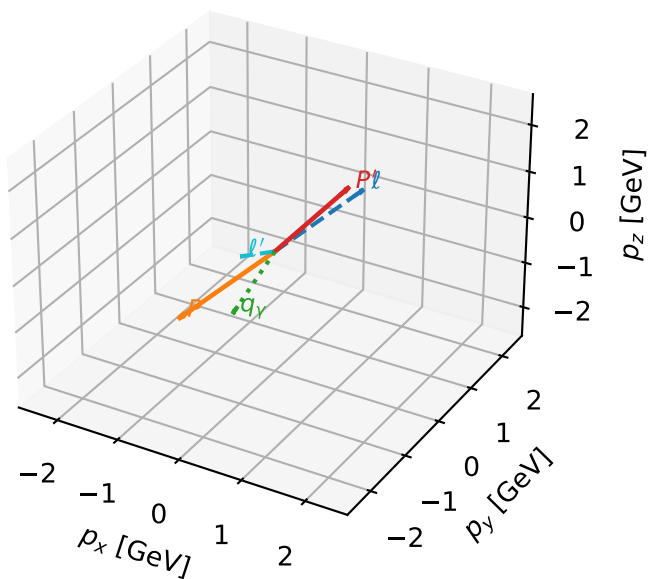
$h_e=+1$ ,  $h_p=+1$ ,  $h_\gamma=-1$   
 massless lepton L, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=99.8%)



# L massless lepton + Tx proton characteristic max $P_{\text{GHZ}}$ + configuration

3D momenta

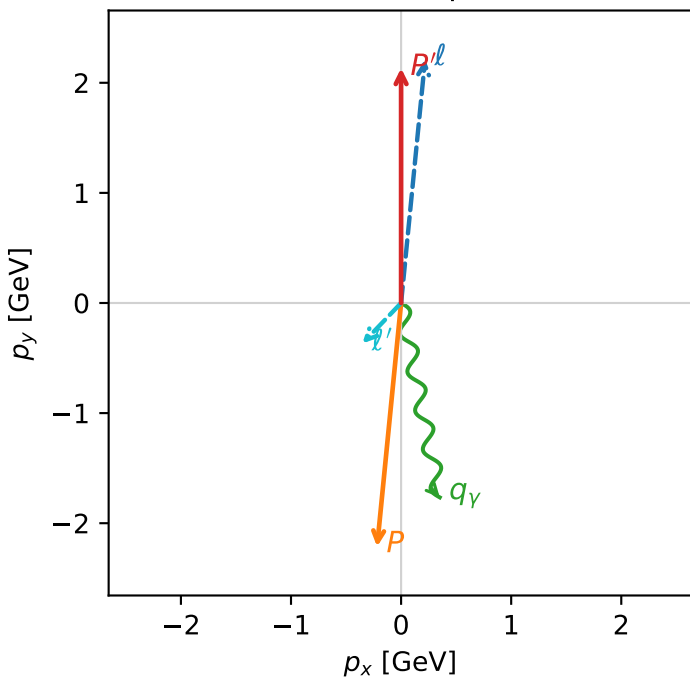


ghz\_purity\_high\_egamma\_ltx\_region\_1  $P_{\text{GHZ}}=0.482787$   
 region: high\_Egamma\_LTx\_region\_1  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_high\_Egamma  
 incoming state:  $|L_{\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.13301$   
 $\theta_{\text{in}}=1.57079$ ,  $\phi_{\ell}=1.47262$ ,  $\phi_P=4.61421$   
 $E_V=1.8$ ,  $\phi_V=4.90874$   
 $Q^2=4.0423$ ,  $x_B=0.202498$ ,  $t=-18.9344$ ,  $W^2=16.7997$ ,  $y=0.967344$   
 $\theta(\ell', \gamma)=54.9402$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 0.2180$   $p_y= 2.2137$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -0.2180$   $p_y= -2.2137$   $p_z= 0.0000$   
 $\ell'$   $E= 0.5084$   $p_x= -0.3512$   $p_y= -0.3676$   $p_z= 0.0000$   
 $P'$   $E= 2.3301$   $p_x= 0.0000$   $p_y= 2.1330$   $p_z= 0.0000$   
 $q_\gamma$   $E= 1.8000$   $p_x= 0.3512$   $p_y= -1.7654$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=99.9%)

